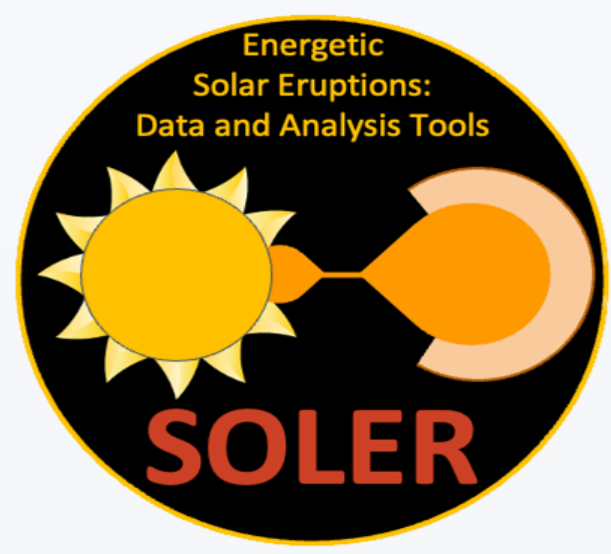




# Enabling Comprehensive Solar Eruption Analyses: Tools developed in the European Horizon Project SOLER



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Observatoire de Paris

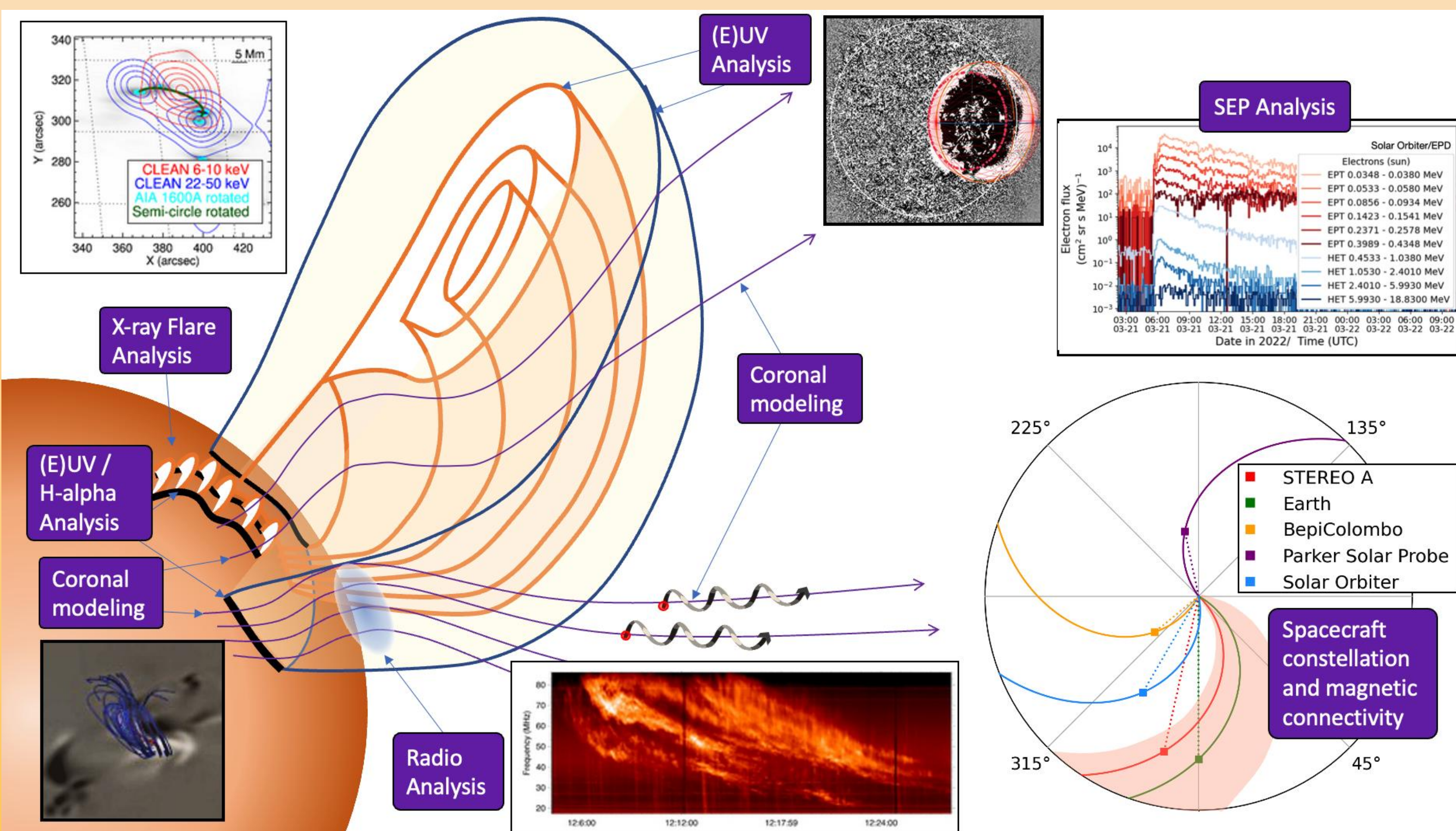
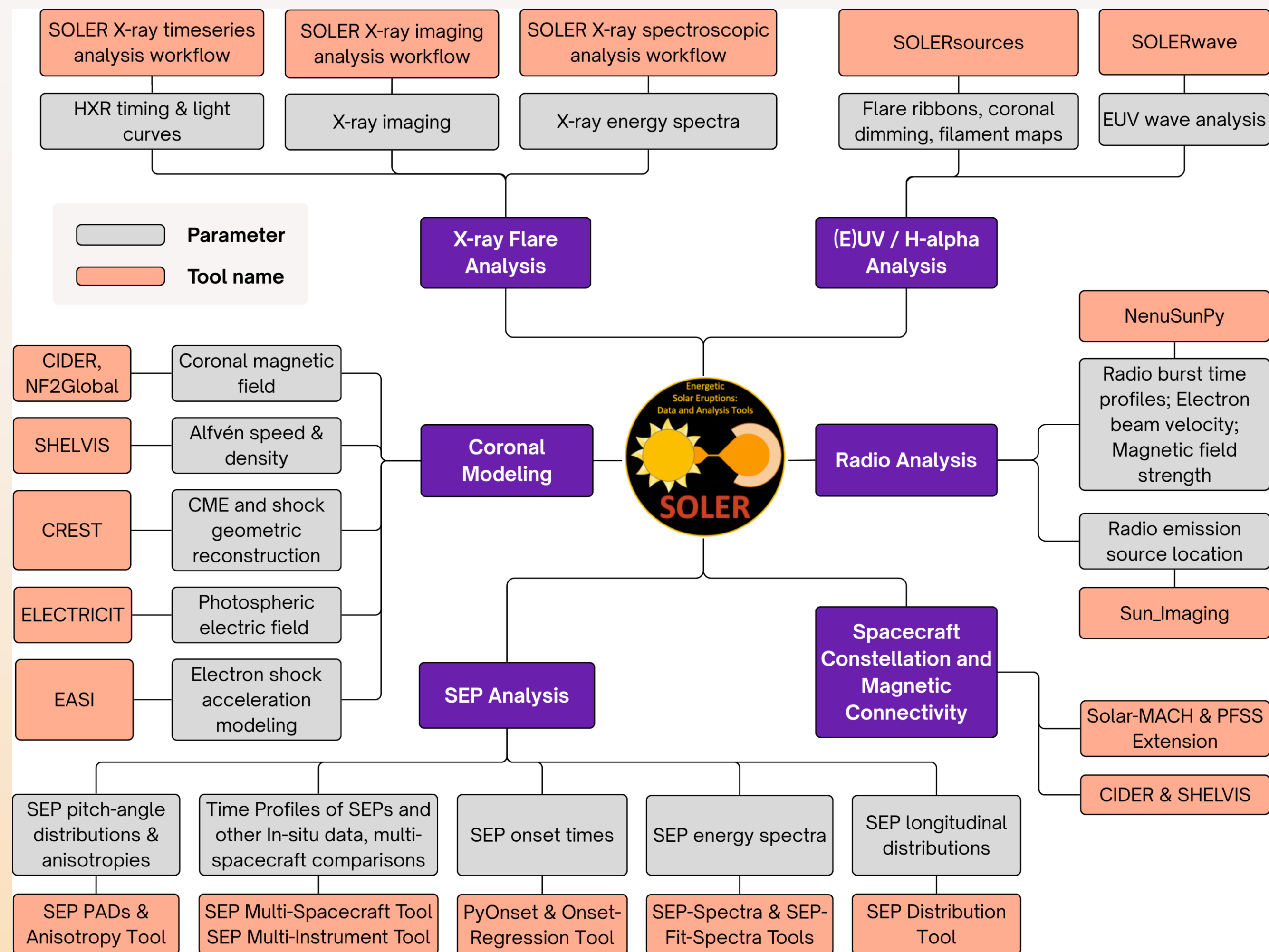


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Solar eruptions comprise a multitude of phenomena such as flares, coronal mass ejections (CMEs), large-scale coronal waves, radio bursts, and energetic particles traveling through interplanetary space. These phenomena are observed with a variety of instrumentation, including remote sensing and in-situ detectors. Typically, a comprehensive analysis of such an event necessitates the expertise of a skilled and broad team.

The "Energetic Solar Eruptions: Data and Analysis Tools" (SOLER) project endeavors to establish a comprehensive analysis platform for the study of solar eruptions. This platform is designed to enable a single user to apply diverse analysis methods and generate research-level results and visualizations. To this end, the project partners developed a multitude of Python-based programs. Each tool is accompanied by a "Plug and Play" Jupyter Notebook, which offers illustrative examples and use cases of the tool.



Starting from the photosphere outward, the tool suite encompasses tools for the analysis of X-ray emissions from flares and the influence of solar energetic events on the corona, manifesting as coronal dimming and large-scale coronal waves (observed in the extreme ultraviolet). Additional tools have been developed to analyze the radio emission created by these events and the solar energetic particles (SEPs) accelerated, observed at different locations in the heliosphere by different space-based missions. Lastly, the tool list includes a coronal modeling tool to determine magnetic connection between SEPs and their possible solar source regions.

The SOLER project will also use the developed tools to investigate events associated with M- and X-class flares from solar cycle no. 25, to provide the community with preprocessed **data products and catalogs**, facilitating future interdisciplinary studies of energetic solar eruptions.

**SOLER Catalogs**  
Downloadable on a stream-lit website allowing for easy prefiltering



**JupyterHub server** makes delivered tools freely available to use without installation.



**GitHub** makes all codes freely available to use and contribute to.



**SOLER Website**  
Information about the project status and workshops



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